

Muhammad Talal Bin Nadeem

+353 870336407 📍 Ireland

✉ talalnadeem24@gmail.com | **in** <https://www.linkedin.com/in/talal-nadeem/> | 🌐 <https://tnz-5.github.io/portfolio/> |
🔗 <https://github.com/TNZ-5>

TECHNICAL SKILLS

Languages: Java, Python, SQL

Frameworks & Libraries: FastAPI, Flask

Databases: MySQL, PostgreSQL

Testing: JUnit, Pytest

DevOps & Cloud Infra: Linux, Maven, Git, Bitbucket, Docker, Kubernetes, AWS, CI/CD (Jenkins, Bitbucket Pipelines)

WORK EXPERIENCE

Infrastructure Engineer

ESSENTIA INC

Jul 2025 - Present

United States, NC

- Engineered end-to-end CI/CD pipelines on AWS EC2 using Jenkins reducing deployment time by **60%** and increasing release reliability across **10+ services**.
- Developed the eSync platform (Cluster Management System) to automate Kubernetes cluster deployment, reducing cluster setup time by 3 hours.
- Provisioned AWS Infrastructure As Code (IaC) using **Pulumi** for AWS resource orchestration & security role
- Owned backend feature development building Restful API in **Python** (FastAPI) for External System Integrations (**15+** FastField Forms) incorporating **AI-powered data processing and automation** to streamline downstream workflows.

Software Engineer II (Backend)

HOTELKEY PVT LTD

Jun 2023 - Jul 2025

United States, TX

- Implemented and Optimised **Java (Spring Boot)** microservices that integrated with Booking.com, Expedia, and Airbnb for real time sync for pricing, availability, and inventory across **7,000+** properties.
- Enhanced service reliability by **25%** through integration, contract, and regression test suites
- Built an automated **Content Publishing service**, reducing update latency from **15 minutes to under 2 minutes** for near real time customer-facing updates.
- Migrated **20+** services and **100K+** lines of legacy code from **Java 8 to Java 21**. Leveraged records, switch expressions, and JVM/runtime optimisations to improve maintainability delivering **15%** reduction in latency and **10%** lower memory utilisation.

EDUCATION

National College Of Ireland

MSc Data Analytics

National University of Computer and Emerging Sciences

Bachelor of Computer Science

Projects:

Chords Compiler | C++, Linux, Kernel

- Applied systems programming (Linux kernel) and low-level software engineering principles to develop a C++ compiler, implementing lexical analysis, parsing, and AST-based representations while exploring process execution, system behaviour, and OS (Operating Systems) level abstractions.
(<https://github.com/TNZ-5/ChordsCompiler>)

AWARDS AND ACCOLADES

TEDxFAST - Curator

2023

National University of Computer and Emerging Sciences

- Curated for TEDxFAST (<https://www.ted.com/tedx/events/52803>) attracting over 300 attendees and featuring 7 speakers, the event aimed to inspire and empower individuals to manifest dreams into reality.

Dean's List High Honor

2024

National University of Computer and Emerging Sciences

- Academic Award For Spring 2024 (<https://www.nu.edu.pk/Campus/Lahore/DeanLists>)